Lab 02: Build a custom VPC with public/private subnets and Deploy EC2 Instances

Part 1: Build a custom VPC with public/private subnets:

In this lab, we will create a custom Virtual Private Cloud (VPC) in AWS with:

. One public subnet for internet-facing resources.

. One private subnet for internal resources.

. Proper routing and security configurations.

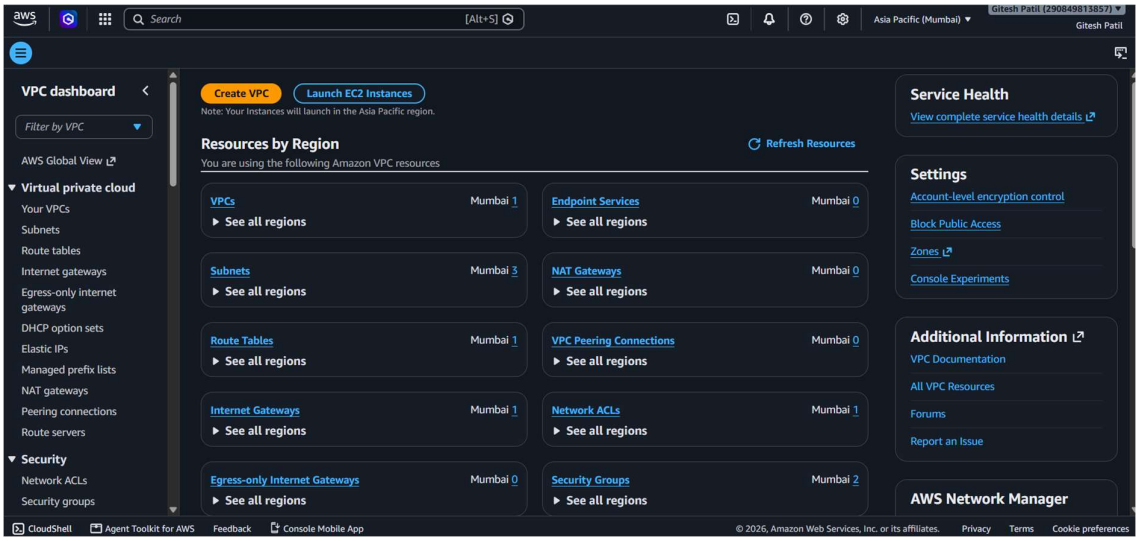
IAM Policy:

```
{
"Version": "2012-10-17",
"Statement": [
{
        "Sid": "VPCManagementPermissions",
        "Effect": "Allow",
        "Action": [
                "ec2:CreateVpc",
                "ec2:DeleteVpc",
                "ec2:DescribeVpcs",
                "ec2: CreateSubnet",
                "ec2:DeleteSubnet",
                "ec2:DescribeSubnets",
                "ec2: CreateInternetGateway",
                "ec2:AttachInternetGateway",
                "ec2:DeleteInternetGateway",
                "ec2:DescribeInternetGateways",
                "ec2: CreateRouteTable",
                "ec2:AssociateRouteTable",
                "ec2: DeleteRouteTable",
                "ec2: DescribeRoutes",
```

```
                "ec2: CreateSecurityGroup",
                "ec2: AuthorizeSecurityGroupIngress"
                "ec2: AuthorizeSecurityGroupEgress",
                "ec2: RevokeSecurityGroupIngress",
                "ec2: RevokeSecurityGroupEgress",
                "ec2: DescribeSecurityGroups"
                ],
                "Resource": "*"
        }
]
```

Step 1: Create VPC

1. Navigate to VPC Console > Create VPC.



2. Select VPC and more option.

3. Configure:

Name: CustomVPC

o IPv4 CIDR: 10.0.0.0/16

o No. of AZs = 3

o No. of Public Subnets = 2

o No. of Private Subnets = 2

4. Expand the Customize subnet CIDR blocks and enter :

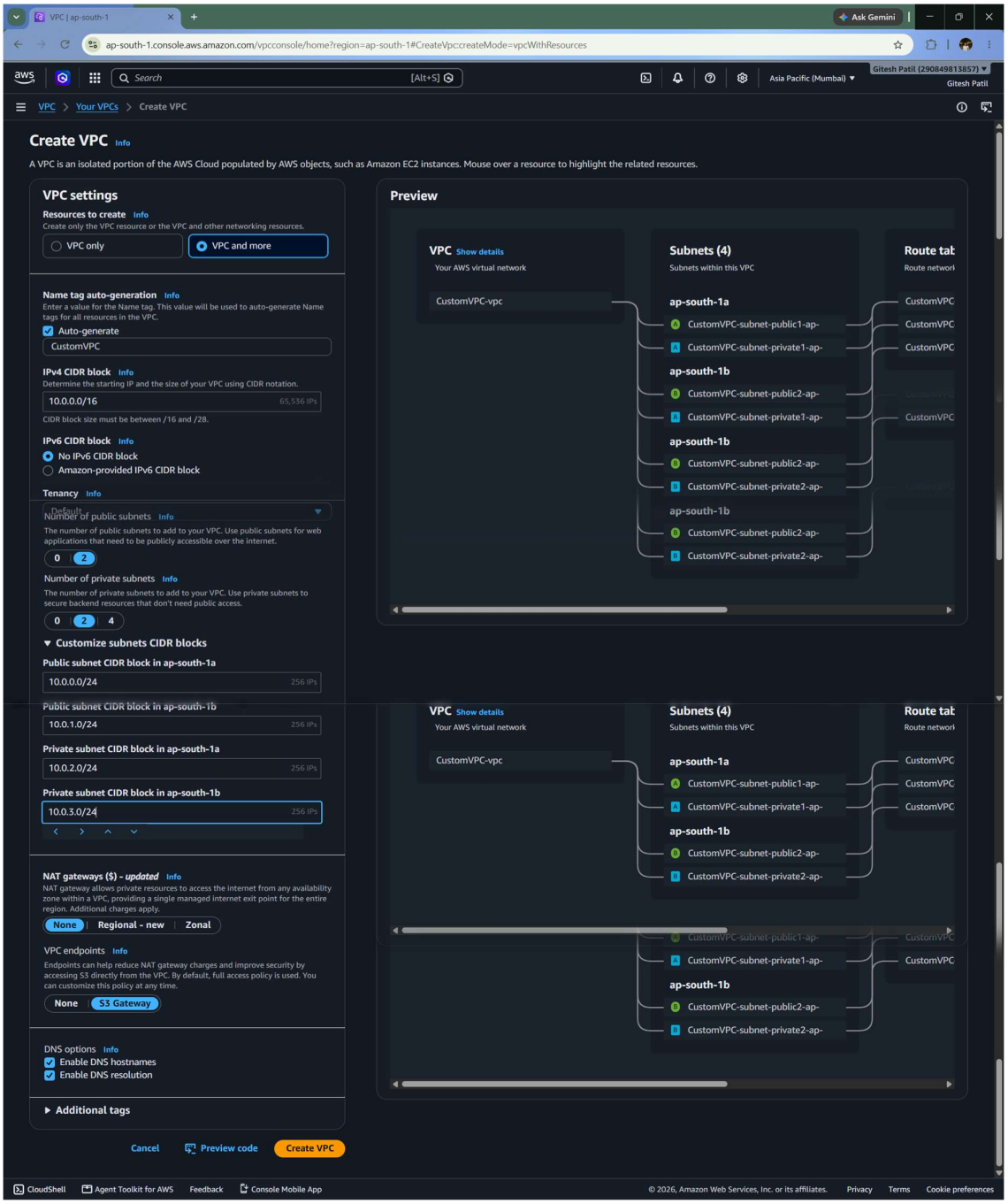
Public Subnet in AZ1 -> 10.0.0.0/24

Public Subnet in AZ2 -> 10.0.1.0/24

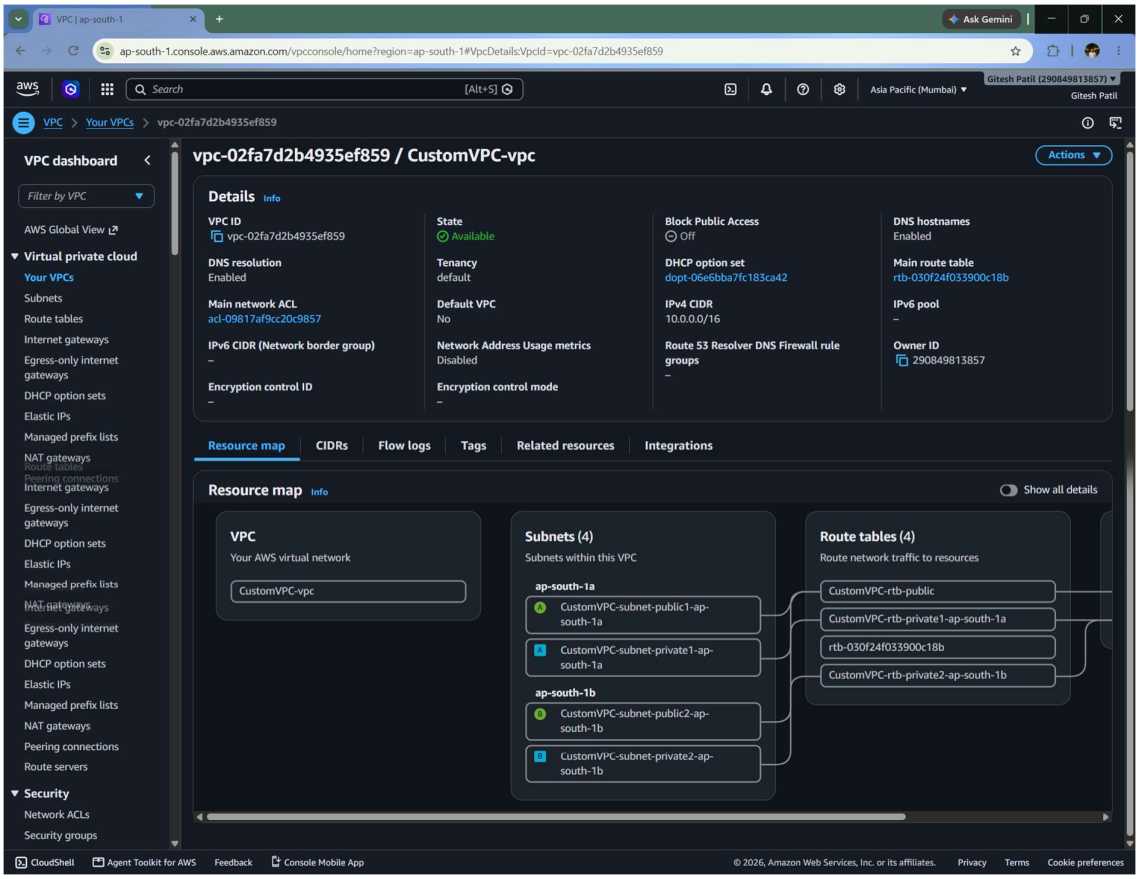
Private Subnet in AZ1 -> 10.0.2.0/24

Private Subnet in AZ2 -> 10.0.3.0/24

Also, verify the traffic flow in the preview on the right and click on Create VPC button at the bottom.



5. Your VPC along with its subcomponents will be created simultaneously. Click on View VPC button to view the VPC details.

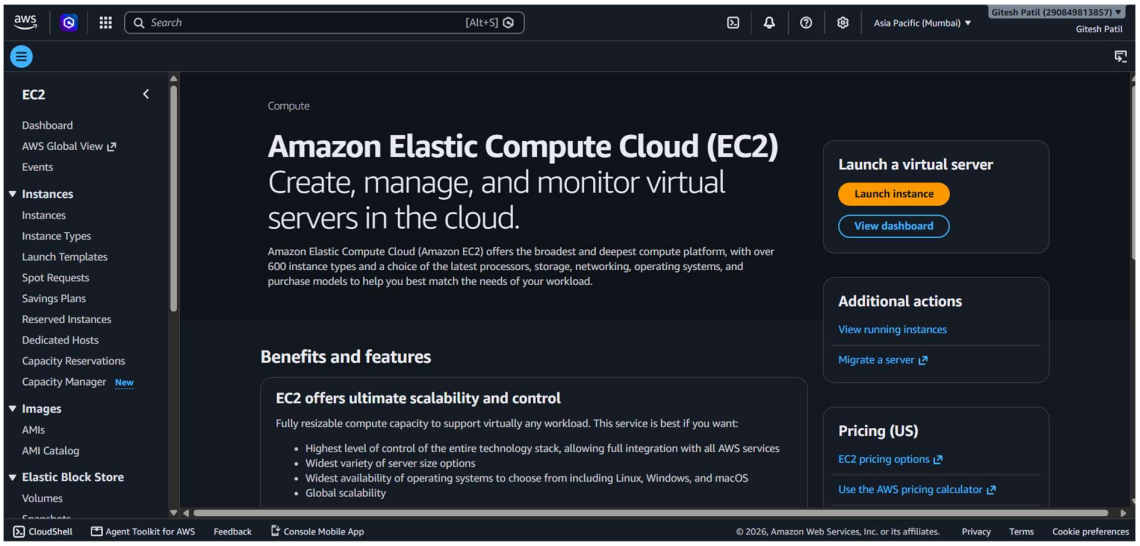


Part 2: Deploy EC2 in private subnet via Bastion Host

In this lab, we will create & deploy an EC2 instance in a private subnet and access it securely through a Bastion Host in the public subnet.

Step 1: Launch a Bastion Host

1. Navigate to EC2 Console -> Launch Instance



2. Configure :

Name: Bastion Host

AMI: Amazon Linux

Instance type: t3.micro

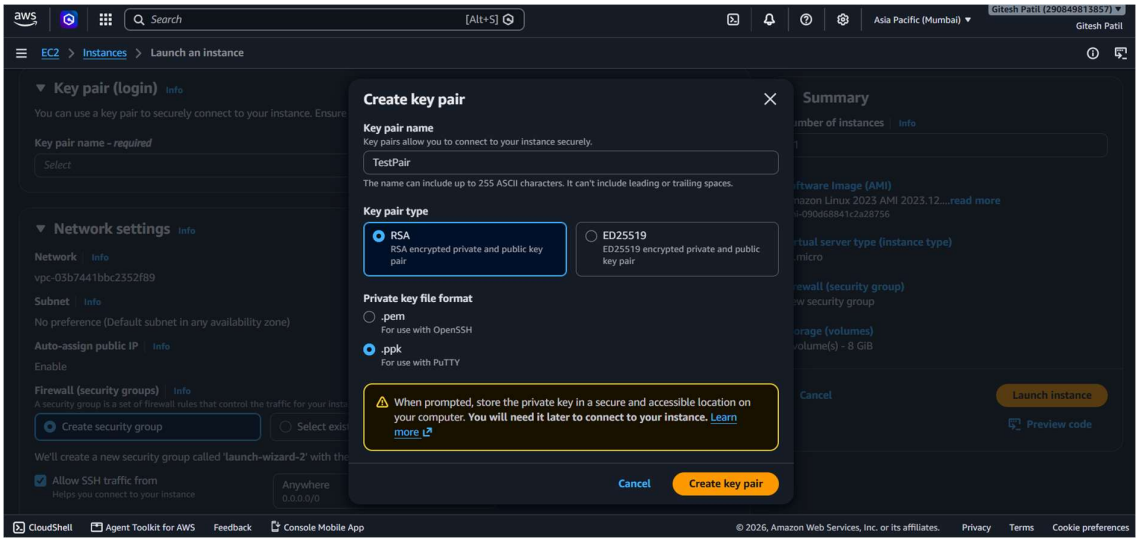
3. In the Key pair (login) section click on Create new key pair and on the subsequent screen configure as follows:

Name: TestPair

Key pair type: RSA

Private key file format: .ppk

After entering these details click on Create key pair button and save the key pair file securely on your local machine.



4. Now, click on the Edit button next to Network settings and configure as follows:

VPC: Custom VPC

Subnet: Select any public subnet

Auto assign Public IP: Enable

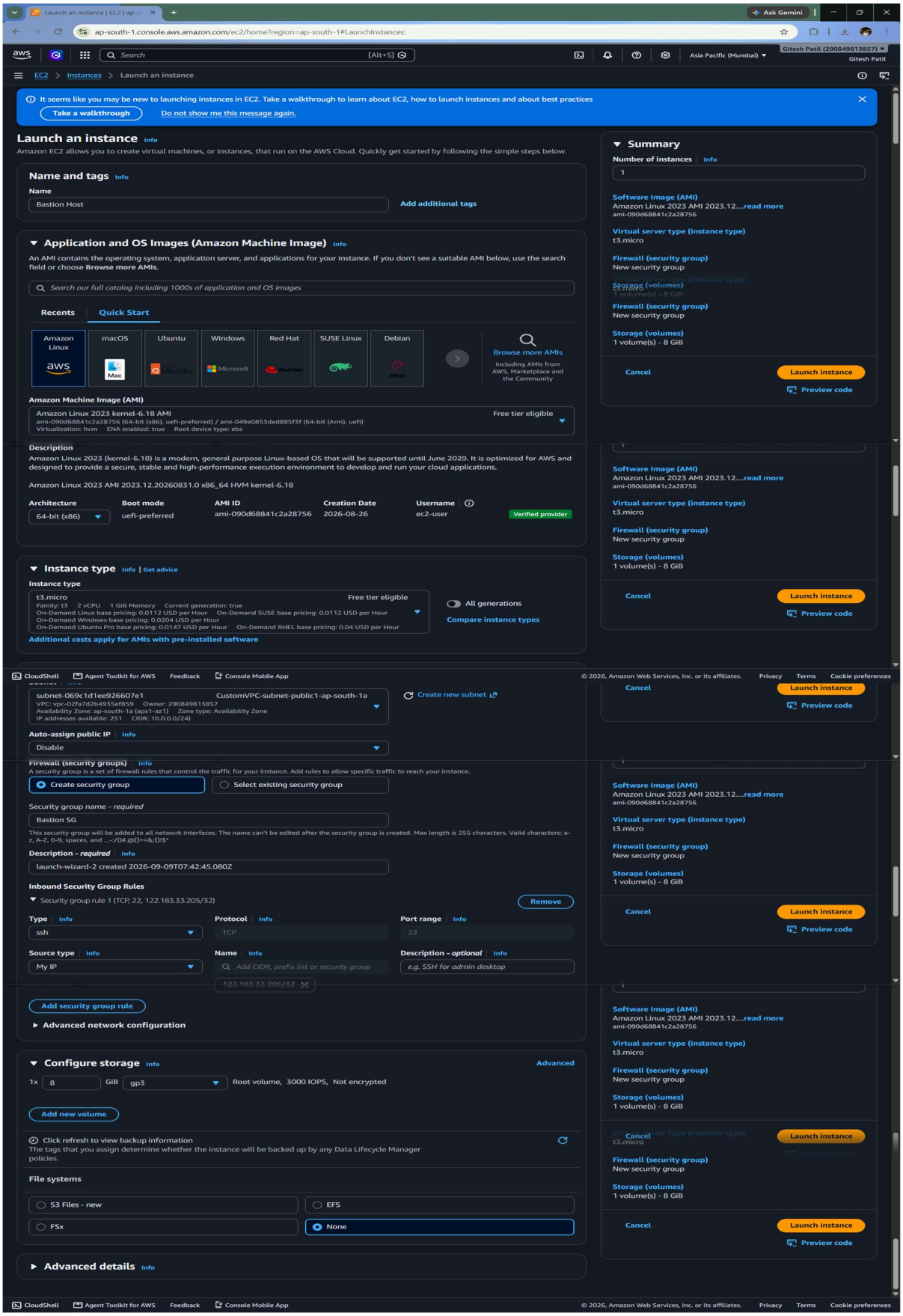
Security group name: Bastion SG

For the Inbound rule:

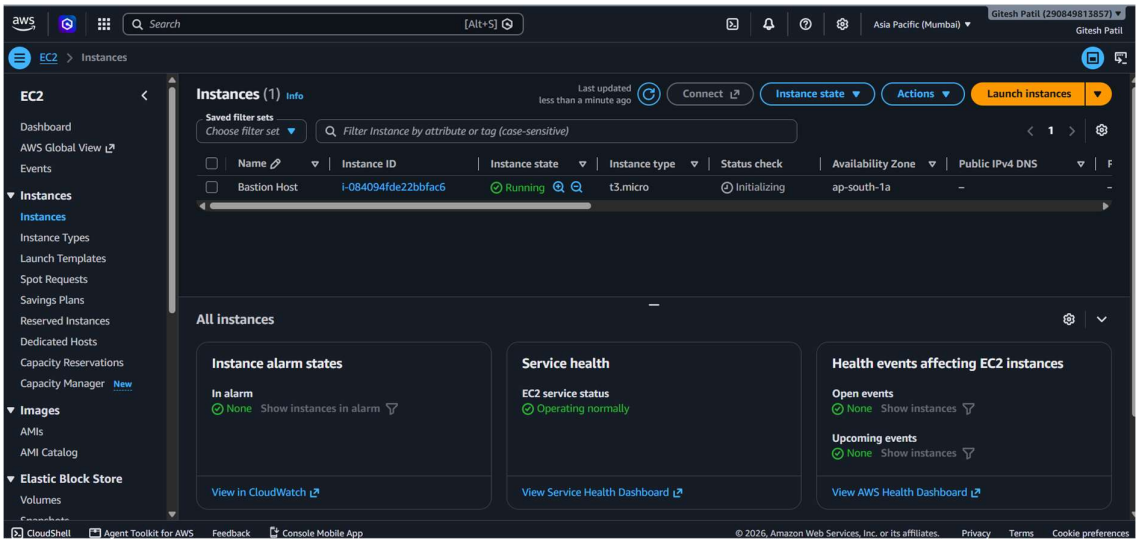
Type: SSH

Source type: My IP

After configuring all this, just click on Launch instance button.


Launch an instance | EC2 | ap-south-1
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Search
[Alt+S]
Asia Pacific (Mumbai)
Gitesh Patil (290849813857)
Gitesh Patil
EC2 > Instances > Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again.
Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
Bastion Host
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
ami-090d68841c2a28756 (64-bit (x86), uefi-preferred) / ami-049e0853ded885f3f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs
Free tier eligible
Description
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20260831.0 x86_64 HVM kernel-6.18
Architecture
64-bit (x86)
Boot mode
uefi-preferred
AMI ID
ami-090d68841c2a28756
Creation Date
2026-08-26
Username
ec2-user
Verified provider
Instance type Info | Get advice
Instance type
t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour
Free tier eligible
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-090d68841c2a28756
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
subnet-069c1d1ee926607e1 CustomVPC-subnet-public1-ap-south-1a
VPC: vpc-02fa7d2b4935ef859 Owner: 290849813857
Availability Zone: ap-south-1a (aps1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.0.0/24)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Security group name - required
Bastion SG
This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and ._-:/()#,@[]+=&;{}!$*
Description - required Info
launch-wizard-2 created 2026-09-09T07:42:45.080Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, 122.183.33.205/32)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
My IP
Name Info
Add CIDR, prefix list or security group
Description - optional Info
e.g. SSH for admin desktop
122.183.33.205/32
Add security group rule
Advanced network configuration
Configure storage Info
Advanced
1x
8
GiB
gp3
Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info

5. Go to the Instances page and you will find your Bastion Host running over there.



Part 3: Launch Private EC2

1. Click on Launch Instance.

2. Configure:

O Name: Private-Bastion Host

o AMI: Amazon Linux

o Instance type: t3.micro

o Key pair: TestPair

3. Now, click on the Edit button next to Network settings and configure as follows:

VPC: Custom VPC

Subnet: Select any public subnet

Auto assign Public IP: Enable

Security group name: Bastion SG

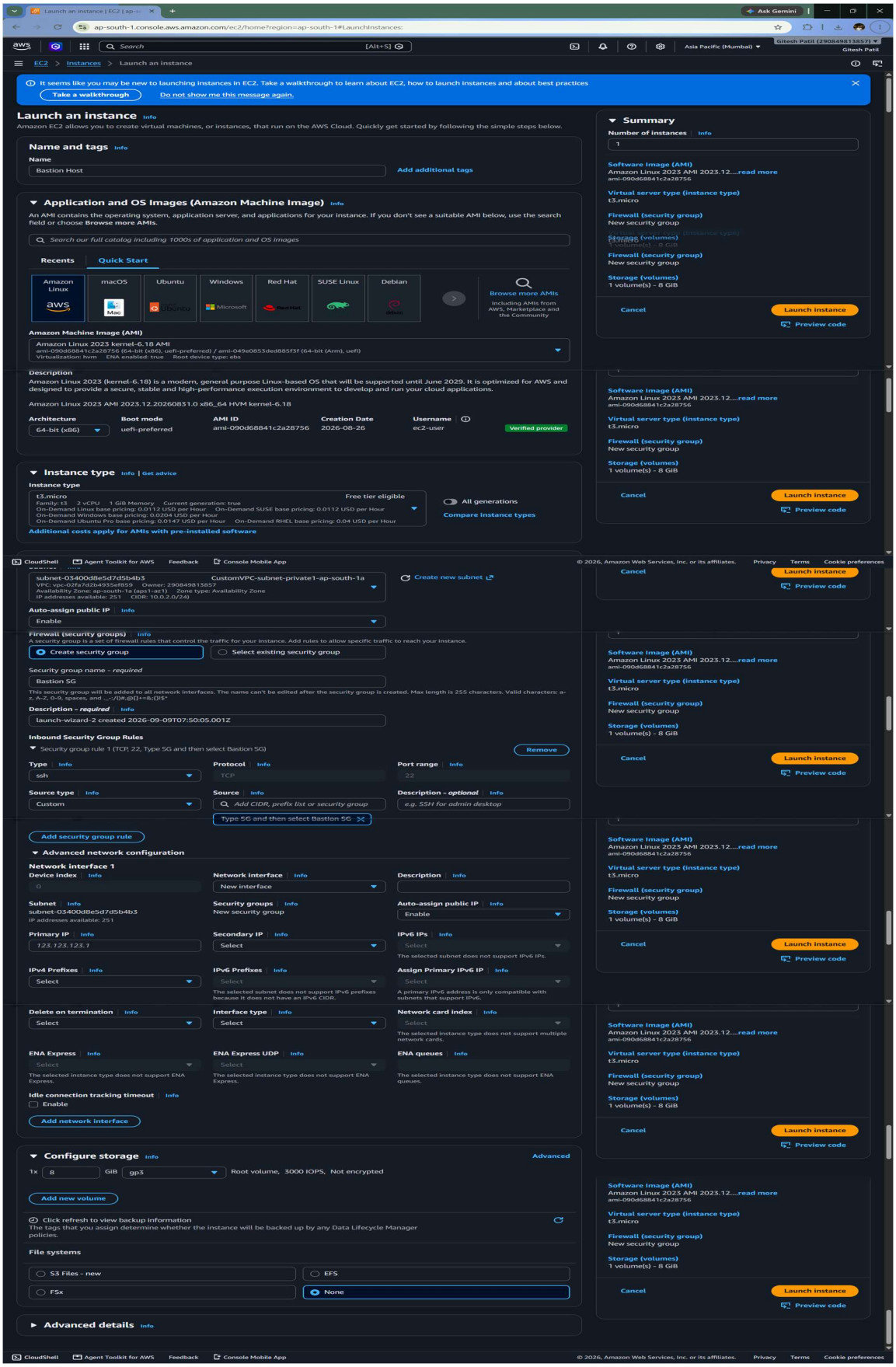
For the Inbound rule:

Type : SSH

Source type : Custom

Source : Type SG and then select Bastion SG

After configuring all this, just click on Launch instance button.


Launch an instance | EC2 | ap-south-1
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
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Cancel
Launch instance
Preview code
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Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
subnet-03400d8e5d7d5b4b3
CustomVPC-subnet-private1-ap-south-1a
VPC: vpc-02fa7d2b4935ef859 Owner: 290849813857
Availability Zone: ap-south-1a (apa1-az1) Zone type: Availability Zone
IP addresses available: 251 CIDR: 10.0.2.0/24)
Create new subnet
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Description - required Info
launch-wizard-2 created 2026-09-09T07:50:05.001Z
Inbound Security Group Rules
Security group rule 1 (TCP, 22, Type SG and then select Bastion SG)
Remove
Type Info
ssh
Protocol Info
TCP
Port range Info
22
Source type Info
Custom
Source Info
Add CIDR, prefix list or security group
Type SG and then select Bastion SG
Description - optional Info
e.g. SSH for admin desktop
Add security group rule
Advanced network configuration
Network interface 1
Device index Info
0
Network interface Info
New interface
Description Info
Subnet Info
subnet-03400d8e5d7d5b4b3
IP addresses available: 251
Security groups Info
New security group
Auto-assign public IP Info
Enable
Primary IP Info
123.123.123.1
Secondary IP Info
Select
IPv6 IPs Info
Select
The selected subnet does not support IPv6 IPs.
IPv4 Prefixes Info
Select
IPv6 Prefixes Info
Select
The selected subnet does not support IPv6 prefixes because it does not have an IPv6 CIDR.
Assign Primary IPv6 IP Info
Select
A primary IPv6 address is only compatible with subnets that support IPv6.
Delete on termination Info
Select
Interface type Info
Select
Network card index Info
Select
The selected instance type does not support multiple network cards.
ENA Express Info
Select
The selected instance type does not support ENA Express.
ENA Express UDP Info
Select
The selected instance type does not support ENA Express.
ENA queues Info
The selected instance type does not support ENA queues.
Idle connection tracking timeout Info
Enable
Add network interface
Configure storage Info
Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info

4. Go to the Instances page and you will find your Private EC2 instance running over there.

